

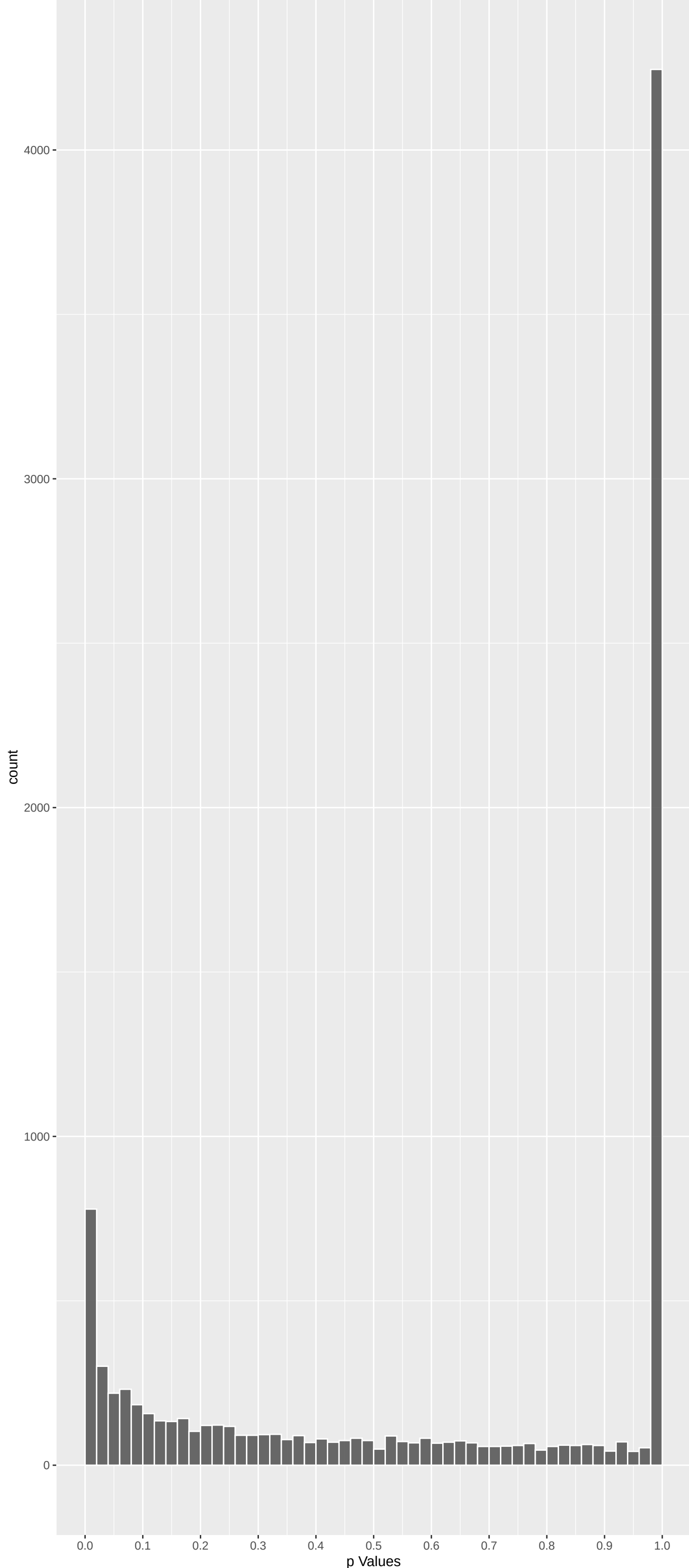
Top genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
BPIFB1	6.257372	1.174557e-09	1.761e-05	2.642e-01
CLDN16	5.586257	6.960465e-08	5.053e-04	1.000e+00
IL1R2	5.470317	1.347693e-07	5.053e-04	1.000e+00
FGD5	5.504803	1.108742e-07	5.053e-04	1.000e+00
CLPS	5.391127	2.100515e-07	5.250e-04	1.000e+00
FAR2	5.393492	2.073041e-07	5.250e-04	1.000e+00
RNF216	5.198525	6.026283e-07	1.291e-03	1.000e+00
NEK5	-5.105738	9.885194e-07	1.853e-03	1.000e+00
PARP12	5.044260	1.365840e-06	2.276e-03	1.000e+00
CPE	4.880561	3.173538e-06	4.759e-03	1.000e+00
GCDH	4.825817	4.182905e-06	4.825e-03	1.000e+00
SCNN1G	4.847044	3.759448e-06	4.825e-03	1.000e+00
SLC23A1	4.839950	3.896151e-06	4.825e-03	1.000e+00
PROM1	4.804689	4.649758e-06	4.981e-03	1.000e+00
GPAT2	4.752485	6.027938e-06	6.027e-03	1.000e+00
CYP11A1	4.735981	6.539948e-06	6.130e-03	1.000e+00
SCAMP2	4.695988	7.959651e-06	7.022e-03	1.000e+00
NSMCE4A	4.658929	9.535746e-06	7.945e-03	1.000e+00
PROC	4.553030	1.586365e-05	8.898e-03	1.000e+00
CAPN7	4.560511	1.530877e-05	8.898e-03	1.000e+00
RBSN	4.577154	1.414030e-05	8.898e-03	1.000e+00
MAST3	4.550968	1.601986e-05	8.898e-03	1.000e+00
CHAC2	4.566589	1.487172e-05	8.898e-03	1.000e+00
GP9	4.578534	1.404735e-05	8.898e-03	1.000e+00
CFAP47	4.591913	1.317603e-05	8.898e-03	1.000e+00

Top genes by Q-Value non-permuted

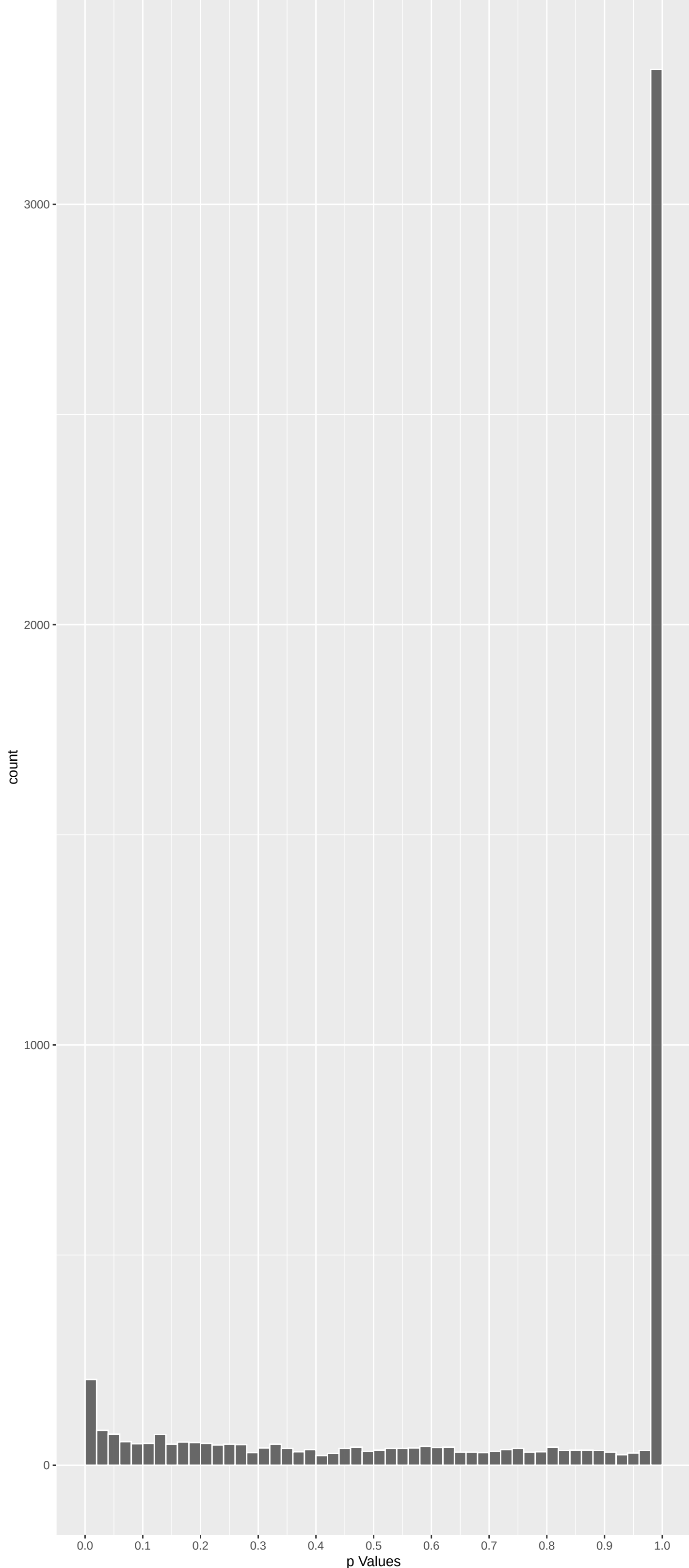
Gene	Rho	P	p.adj	qValueNoperm
BPIFB1	6.2573719	1.174557e-09	1.761e-05	2.642e-01
M6PR	0.2739266	1.000000e+00	1.000e+00	1.000e+00
FKBP4	1.9484592	1.540801e-01	8.461e-01	1.000e+00
CYP26B1	-0.1086069	1.000000e+00	1.000e+00	1.000e+00
NDUFAF7	-0.1958252	6.952943e-01	1.000e+00	1.000e+00
FUCA2	-0.3323291	1.000000e+00	1.000e+00	1.000e+00
DBNDD1	0.4947402	1.000000e+00	1.000e+00	1.000e+00
HS3ST1	1.4133227	4.726826e-01	1.000e+00	1.000e+00
SEMA3F	0.8133988	1.000000e+00	1.000e+00	1.000e+00
CFTR	-1.0458077	8.869494e-01	1.000e+00	1.000e+00
CYP51A1	1.5223834	3.837391e-01	1.000e+00	1.000e+00
USP28	-0.5717125	1.000000e+00	1.000e+00	1.000e+00
SLC7A2	-0.0166267	1.000000e+00	1.000e+00	1.000e+00
HSPB6	0.5525808	1.000000e+00	1.000e+00	1.000e+00
PKD4	3.5963829	9.680181e-04	6.344e-02	1.000e+00
USH1C	-0.4747736	1.000000e+00	1.000e+00	1.000e+00
BAIAP2L1	1.8132874	2.093625e-01	9.772e-01	1.000e+00
TMEM132A	0.5624819	1.000000e+00	1.000e+00	1.000e+00
DVL2	-2.7421397	1.831211e-02	2.940e-01	1.000e+00
RPAP3	-0.6391680	1.000000e+00	1.000e+00	1.000e+00
HOXA11	-0.9521654	1.000000e+00	1.000e+00	1.000e+00
TRAPPC6A	1.0053199	9.442286e-01	1.000e+00	1.000e+00
SPATA20	-0.2077876	1.000000e+00	1.000e+00	1.000e+00
CD79B	0.7132749	1.000000e+00	1.000e+00	1.000e+00
RHBDD2	3.3740023	2.222512e-03	9.845e-02	1.000e+00

Positive Rho Non-permuted



Top Positive genes by P-value non-permuted

Negative Rho Non-permuted



Top Negative genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
BPIFB1	6.257372	1.174557e-09	1.761e-05	2.642e-01
CLDN16	5.586257	6.960465e-08	5.053e-04	1.000e+00
IL1R2	5.470317	1.347693e-07	5.053e-04	1.000e+00
FGD5	5.504803	1.108742e-07	5.053e-04	1.000e+00
CLPS	5.391127	2.100515e-07	5.250e-04	1.000e+00
FAR2	5.393492	2.073041e-07	5.250e-04	1.000e+00
RNF216	5.198525	6.026283e-07	1.291e-03	1.000e+00
PARP12	5.044260	1.365840e-06	2.276e-03	1.000e+00
CPE	4.880561	3.173538e-06	4.759e-03	1.000e+00
GCDH	4.825817	4.182905e-06	4.825e-03	1.000e+00

Gene	Rho	P	p.adj	qValueNoperm
NEK5	-5.105738	9.885194e-07	1.853e-03	1.000e+00
GPR179	-4.587791	1.343881e-05	8.898e-03	1.000e+00
BIRC3	-4.459560	2.463841e-05	1.073e-02	1.000e+00
FAT1	-4.350401	4.076662e-05	1.359e-02	1.000e+00
ZNF311	-4.243873	6.590846e-05	1.830e-02	1.000e+00
SNRPD3	-4.215000	7.493355e-05	1.972e-02	1.000e+00
PLPP1	-4.123987	1.117108e-04	2.359e-02	1.000e+00
IQGAP3	-4.116989	1.151561e-04	2.359e-02	1.000e+00
DISP1	-4.131148	1.082866e-04	2.359e-02	1.000e+00
H2BC21	-4.060856	1.466796e-04	2.600e-02	1.000e+00

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Inorganic Cation Transmembrane Transport	0.12350814	268	5.289e-12	2.849e-08	SCNN1G:10 SLC6A6:30 ANO6:50 ATP6V0A2:62 SLC39A13:127 ATP2A1:137
Monoatomic Cation Transmembrane Transport	0.11555125	265	1.184e-10	3.189e-07	SCNN1G:10 SLC6A6:30 ANO6:50 ATP6V0A2:62 SLC39A13:127 ATP2A1:137
Nitrogen Compound Transport (GO:0071705)	0.13351946	144	3.475e-08	6.237e-05	SCAMP2:16 SLC6A6:30 SLC26A4:47 SLC26A8:69 SLC22A4:98 SLC25A11:113
Monocarboxylic Acid Transport (GO:001571)	0.18066230	61	1.089e-06	1.466e-03	UGT1A6:89 SLC6A12:147 SLC10A2:155 SLC22A13:166 FABP1:204 SLC16A5:255
Organic Hydroxy Compound Transport (GO:0)	0.23348579	35	1.774e-06	1.592e-03	SLC10A2:155 SLC22A1:184 SLC01B1:279 ABCD3:285 SLC27A5:501 SLC10A3:617
Sodium Ion Transmembrane Transport (GO:0)	0.15916530	76	1.661e-06	1.592e-03	SCNN1G:10 SLC6A6:30 SLC6A12:147 SCNN1B:162 SLC6A14:224 TRFMA:225
Carboxylic Acid Transport (GO:0046942)	0.19101724	49	3.801e-06	9.924e-03	SLC6A6:30 SLC6A12:147 SLC6A14:224 SLC6A19:244 SLC16A5:255 SLC36A2:283
Cytoplasmic Translation (GO:0002181)	-0.19017812	45	1.031e-05	6.939e-03	RPS23:15 UBA52:19 RPS3A:22 RPL34:23 RACK1:33 RPS18:35
Amino Acid Import Across Plasma Membrane	0.24799381	25	1.782e-05	1.066e-02	SLC6A6:30 SLC22A4:98 SLC6A14:224 SLC1A1:362 SLC1A6:363 SLC7A8:443
Sodium Ion Transport (GO:0006814)	0.13471911	82	2.544e-05	1.370e-02	SCNN1G:10 SLC6A6:30 SLC6A12:147 SCNN1B:162 SLC6A14:224 SLC6A19:244
Fatty Acid Metabolic Process (GO:0006631)	0.11318133	108	5.004e-05	2.310e-02	GPAT2:14 LPIN2:111 ABCB11:181 GPX4:185 ABCD3:285 CPT1A:294
Neutrial Amino Acid Transport (GO:0015804)	0.20375991	33	5.147e-05	2.310e-02	SLC6A6:30 SLC6A14:224 SLC36A2:283 SLC43A1:304 SLC1A1:362 SLC7A8:443
Endomembrane System Organization (GO:001)	-0.09454798	150	6.741e-05	2.420e-02	OBSL1:71 VPS13B:83 SPTBNS:96 CDC42:115 RAB1A:204 TJAP1:309
Inorganic Cation Import Across Plasma Me	0.12021833	93	6.338e-05	2.420e-02	SCNN1G:10 KCNJ1:77 SCNN1B:162 CLU:190 TRPM4:225 SLC9A4:261
Phosphatidylcholine Metabolic Process (G	0.17131551	46	5.897e-05	2.420e-02	PNLIPRP2:82 TMEM86B:321 ENPP2:230 LPCAT3:380 MFSD2A:389 CHPT1:474
Acid Secretion (GO:0046717)	0.39936459	8	9.174e-05	3.088e-02	UGT1A6:89 DRD3:326 SLC51A:675 SLC51B:920 GHRL:1061 SLC9A3R1:2227
Organic Anion Transport (GO:0015711)	0.12361165	80	1.353e-04	4.165e-02	SLC22A13:166 SLC4A1:275 SLC01B1:279 CPTP:352 SLC4A7:379 SLC35B3:427
Organic Cation Transport (GO:0015695)	0.19778914	31	1.392e-04	4.165e-02	SLC22A4:98 SLC6A12:147 SLC22A1:184 SLC5A7:507 SLC6A3:766 SLC22A3:973
Inorganic Anion Transport (GO:0015698)	0.14785601	55	1.510e-04	4.281e-02	SLC26A4:47 ANO6:50 SLC26A8:69 SLC25A11:113 ANO6:262 SLC4A1:275
L-serine Metabolic Process (GO:0006563)	0.34195294	10	1.810e-04	4.874e-02	SDS:303 SRR:842 PSPH:1003 CBS:1047 SERINC3:1619 SDSL:1803
Peptide Biosynthetic Process (GO:0043043)	-0.10432526	105	2.275e-04	5.835e-02	RPS23:15 RPS3A:22 RPL34:23 RPS18:35 RPS27A:37 RPS17:57
Fatty Acid Transport (GO:0015908)	0.17709294	35	2.904e-04	7.019e-02	SLC22A1:184 FABP1:204 MFSD2A:389 CPT1B:773 SLC27A1:791 FABP9:833
Quaternary Ammonium Group Transport (GO:0)	0.30145880	12	2.998e-04	7.019e-02	SLC22A4:98 SLC22A1:184 SLC5A7:507 SLC22A3:973 SLC44A4:1878 SLC44A1:1994
Amino Acid Transmembrane Transport (GO:0)	0.16148219	41	3.497e-04	7.847e-02	SLC6A6:30 SLC22A4:98 SLC6A14:224 SLC26A2:283 SLC1A6:363 SLC7A8:443
Golgi Organization (GO:0007030)	-0.10502031	93	4.755e-04	9.630e-02	OBSL1:71 VPS13B:83 SPTBNS:96 CDC42:115 RAB1A:204 TJAP1:309
Alpha-Amino Acid Biosynthetic Process (G	0.11318133	10	4.828e-04	9.630e-02	BCAT2:126 SDS:303 CPS1:685 ILVBL:937 SDSL:1803 OTC:1866
Regulation Of DNA-templated Transcription	-0.02835730	1447	4.707e-04	9.630e-02	UBE2I:40 SF3B5:77 BRCA1:76 GMMN:78 HSFA:91 GON4L:95
Amino Acid Transport (GO:0006865)	0.15505614	42	5.117e-04	9.840e-02	SLC6A6:30 SLC6A12:147 SLC6A14:224 SLC6A19:244 SLC36A2:283 SLC43A1:304
Alanine Transport (GO:0032328)	0.28752577	12	5.640e-04	9.937e-02	SLC6A6:30 SLC6A14:224 SLC36A2:283 SLC43A1:304 SLC38A2:810 SLC7A10:1035
Negative Regulation Of Viral Genome Repl	0.17883618	31	5.721e-04	9.937e-02	MORC2:40 TASOR:472 EIF2AK2:452 IFIT1:761 PNPPK:1057 BTBD17:1101
Potassium Ion Transmembrane Transport (G	0.08823017	129	5.565e-04	9.937e-02	KCNJ1:77 KCNQ4:174 KCNN4:209 SLC4A4:261 LRRP5:278 KCNK16:334
Regulation Of Transcription By RNA Polym	-0.02743635	1500	5.909e-04	9.943e-02	ACTR2:31 UBED21:34 UBE2I:40 UBE2D:345 AH1:60 RPL17:66
Macromolecule Biosynthetic Process (GO:0)	-0.08736633	128	6.615e-04	9.951e-02	RPS23:15 RPS3A:22 RPL34:23 RPS18:35 RPS27A:37 RPS17:57
Monoatomic Ion Transport (GO:0006811)	0.09170171	102	7.130e-04	9.951e-02	SLC26A4:47 ANO6:50 AQP11:131 SLC9A4:261 SLC4A1:275 SLC36A2:283
Negative Regulation Of Autophagosome Assb	-0.32836220	9	6.473e-04	9.951e-02	SCFD1:69 FEZ2:75 PHF23:1021 NUPR1:1115 MTM1:1465 TMEM39A:1570
Negative Regulation Of Organelle Assembly	-0.17017912	33	7.207e-04	9.951e-02	TBC1D30:67 SCFD1:69 FEZ2:75 TCHP:649 PFH23:1021 CEP97:1026
One-Carbon Compound Transport (GO:001975)	0.18218762	29	6.881e-04	9.951e-02	AQP11:131 SLC4A1:275 SLC4A7:379 SLC39A8:592 HBE1:654 C4A:686
Sodium Ion Import Across Plasma Membrane	0.22682695	19	6.215e-04	9.951e-02	SCNN1G:10 SCNN1B:162 TRPM4:225 SLC9A4:261 SLC9A3:640 SCNN1A:1303
Tight Junction Assembly (GO:0120192)	0.13923044	50	6.666e-04	9.951e-02	CLDN16:2 LSR:36 CLDN18:116 OCELL:249 CDH5:786 PECAM1:815
Inorganic Anion Transmembrane Transport	0.12701851	59	7.489e-04	1.008e-01	ANO6:50 SLC25A11:113 ANO8:262 SLC4A1:275 SLC1A1:362 SLC12A9:709

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.15572906	233	3.123e-16	2.026e-12	SLC6A6:30 SLC26A4:47 SLC22A4:98 CALM1:114 SLC6A12:147 SLC22A1:184
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.09313962	624	3.125e-15	1.014e-11	SCNN1G:10 SLC6A6:30 LSR:36 ATP13A1:43 SLC26A4:47 ANO6:50
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS_	0.20926888	101	3.870e-13	8.369e-10	SLC6A6:30 SLC26A4:47 CALM1:114 SLC6A12:147 SLC6A14:224 SLC6A19:244
HSIAO_LIVER_SPECIFIC_GENES	-0.14788997	196	1.049e-12	1.702e-09	PROC:24 LSR:36 TM4SF4:46 AGXT:86 GYS2:105 LPIN2:111
REACTOME_OXIDATIVE_STRESS_INDUCED_SENESC	-0.21952385	77	2.830e-11	3.672e-08	H2BC21:10 UBA52:19 RPS27A:37 H3C13:39 H4C2:56 H4C14:152
REACTOME_MEIOTIC_RECOMBINATION	-0.27053786	48	9.049e-11	9.784e-08	H2BC21:10 H3C13:39 H4C2:56 BRCA1:76 MLH3:134 H4C14:152
REACTOME_ASSEMBLY_OF_THE_ORC_COMPLEX_AT_	-0.33407584	31	1.224e-10	1.134e-07	H2BC21:10 H3C13:39 H4C2:56 H4C14:152 H4C15:266 H4C3:271
REACTOME_METABOLISM_OF_LIPIDS	0.07258898	648	3.910e-10	3.171e-07	FAR2:5 GPAT2:14 CYP11A1:15 ACER1:29 EHHADH:32 MORC2:40
REACTOME_SIRT1_NEGATIVELY_REGULATES_RRNA	-0.30435775	33	1.452e-09	9.665e-07	H2BC21:10 H3C13:39 H4C2:56 H4C14:152 H4C15:266 H4C3:271
REACTOME_ERCC6_CSB_AND_EHMT2_G9A_POSITIV	-0.1278746	39	1.490e-09	9.665e-07	H2BC21:10 H3C13:39 H4C2:56 H4C14:152 ATAD2B:212 H4C15:266
REACTOME_ASSEMBLY_OF_ANTI_OXIDANT_HOX_GENE	-0.21128746	68	1.730e-09	1.020e-06	H2BC21:10 POLR21:13 H3C13:39 H4C2:56 POLR21:72 H4C14:152
REACTOME_PRC2_METHYLATES_HISTONES_AND_DN	-0.27701941	38	3.471e-09	1.732e-06	H2BC21:10 H3C13:39 H4C2:56 H4C14:152 H4C15:266 H4C3:271
WOO_LIVER_CANCER_RECURRENCE_DN	0.21047270	66	3.428e-09	1.732e-06	GCDH:12 MTPP:146 PLPP3:290 CRYL1:312 SLC2A9:433 CYB5A:489
REACTOME_RNA_POLYMERASE_I_PROMOTER_ESCAP	-0.24687615	47	4.817e-09	2.232e-06	H2BC21:10 H3C13:39 H4C2:56 H4C14:152 H4C15:266 H4C3:271
REACTOME_ACTIVATED_PKN1_STIMULATES_TRANS	-0.31245428	29	5.787e-09	2.503e-06	H2BC21:10 H3C13:39 H4C2:56 H4C14:152 H4C15:266 H4C3:271
REACTOME_DNA_METHYLATION	-0.30057555	30	1.220e-08	4.946e-06	H2BC21:10 H3C13:39 H4C2:56 H4C14:152 H4C15:266 H4C3:271
REACTOME_RECOGNITION_AND_ASSOCIATION_OF_	-0.31470332	27	1.520e-08	5.802e-06	H2BC21:10 H4C2:56 NEIL:397 H4C14:152 H4C15:266 H4C3:271
REACTOME_HDACS_DEACETYLATE_HISTONES	-0.22643216	51	2.249e-08	8.105e-06	H2BC21:10 H3C13:39 H4C2:56 H4C14:152 ATAD2B:212 H4C15:266
GEORGES_TARGETS_OF_MIR192_AND_MIR215	-0.05921119	751	4.380e-08	1.496e-05	ACTR2:31 UBE2D1:34 WDCP:36 LIMCH1:42 SNRPD1:52 BRCA1:76
SANSOM_APC_TARGETS_DN	0.09212460	297	5.224e-08	1.694e-05	VAU1:34 CMBL:49 ACP5:66 ENTPD5:70 SLC22A4:98 PRXL2:100
REACTOME_BIOLOGICAL_OXIDATIONS	0.13039087	145	6.238e-08	1.927e-05	CYP11A1:15 CHAC2:22 CMBL:49 SULT1E1:61 UGT1A6:89 COMT:120
REACTOME_INHIBITION_OF_DNA_RECOMBINATION	-0.28950982	29	6.845e-08	2.018e-05	H2BC21:10 POLR21:13 H4C2:56 POLR23:72 H4C14:152 H4C15:266
REACTOME_HATS_ACETYLATE_HISTONES	-0.16339923	89	1.018e-07	2.751e-05	H2BC21:10 H3C13:39 H4C2:56 ELP2:102 H4C14:152 ELP1:197
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_S	-0.20048672	59	1.017e-07	2.751e-05	H2BC21:10 POLR21:13 H3C13:39 RAN:49 H4C2:56 POLR23:72
REACTOME_TRANSPORT_OF_BILE_SALTS_AND_ORG	0.16731044	82	1.664e-07	4.318e-05	SLC6A6:30 SLC22A4:98 SLC6A12:147 SLC22A1:184 SLC6A14:224
REACTOME_RNA_POLYMERASE_I_TRANSCRIPTION	-0.18689807	65	1.909e-07	4.763e-05	H2BC21:10 H3C13:39 H4C2:56 H4C14:152 ATAD2B:212 H4C15:266
REACTOME_MEIOSIS	-0.17100037	75	3.110e-07	7.204e-05	H2BC21:10 H3C13:39 UBE2I:40 H4C2:56 BRCA1:76 MLH3:134
REACTOME_AMINO_ACID_TRANSPORT_ACROSS_THE	0.26565543	31	3.082e-07	7.204e-05	SLC6A6:30 SLC6A12:147 SLC6A14:224 SLC6A19:244 SLC36A2:283 SLC43A1:304
REACTOME_POSITIVE_EPIGENETIC_REGULATION_	-0.19061807	60	3.323e-07	7.433e-05	H2BC21:10 H3C13:39 H4C2:56 H4C14:152 ATAD2B:212 H4C15:266
REACTOME_CONDENSATION_OF_PROPHASE_CHROMO	-0.24540555	36	3.502e-07	7.440e-05	H2BC21:10 H3C13:39 H4C2:56 H4C14:152 H4C15:266 H4C3:271
REACTOME_BASE_EXCISION_REPAIR_ASI_SITE_PO	-0.25236055	34	3.555e-07	7.440e-05	H2BC21:10 H4C2:56 NEIL:397 H4C14:152 H4C15:266 H4C3:271
REACTOME_SLC_TRANSPORTER_DISORDERS	0.15525598	89	4.233e-07	8.580e-05	SLC26A4:47 NUP210:215 SLC6A14:224 SLC6A19:244 SLC4A1:275 SLC01B1:279
NIKOLSKY_BREAST_CANCER_5P15_AMPLICON	0.32385625	20	5.343e-07	1.050e-04	SLC6A19:244 AHRP:277 SDHA:333 SLC9A3:640 SLC6A3:766 PPPP:828
REACTOME_B_WICH_COMPLEX_POSITIVELY_REGUL	-0.20973831	47	6.598e-07	1.259e-04	H2BC21:10 H3C13:39 H4C2:56 H4C14:152 H4C15:266 H4C3:271
CAIRO_NEOPLASTASTOMA_DN	0.09490107	231	7.167e-07	1.328e-04	GCDH:12 AGXT:86 GYS2:105 LPIN2:111 ABCB11:181 SLC22A1:184
WP_NRF2_PATHWAY	0.13549235	112	7.497e-07	1.351e-04	SLC6A6:30 UGT1A6:89 SLC39A13:127 SLC6A14:224 MGST3:227 TXNRD3:234
REACTOME_FORMATION_OF_THE_BETA_CATENIN_T	-0.20132392	50	8.531e-07	1.419e-04	H2BC21:10 H3C13:39 H4C2:56 H4C14:152 H4C15:266 H4C3:271
WP_TRANSULFURATION_ONECARBON_METABOLISM	0.18396078	60	8.398e-07	1.419e-04	BCAT2:126 BHTM:164 GPX4:185 GNTM:273 PHGDH:293 CHPT1:474
WEBER_METHYLATED_HCP_IN_FIBROBLAST_DN	0.24449674	34	8.111e-07	1.419e-04	DKKL1:59 TRPM4:225 TULP2:370 DAZL:481 RBMXL:2.485 ACRBP:486
WP_PROXIMAL_TUBULE_TRANSPORT	0.20567570	47	1.080e-06	1.751e-04	ATP6V0A4:148 SLC6A19:244 SLC36A2:283 AQP1:289 SLC1A1:362 SLC7A8:443

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Liver neoplasms	0.05887179	1017	4.930e-10	4.835e-06	PROM1:13 PROC:24 SUGP1:58 SULT1E1:61 ATG7:84 UGT1A6:89
Cystic Fibrosis	0.07888134	419	3.919e-08	1.922e-04	BP1FB1:1 SCNN1G:10 SSGM3:39 SULT1E1:61 ST14:99 CALM1:114
Hypertensive disease	0.04747535	1069	2.875e-07	7.049e-04	CLDN16:2 SCNN1G:10 CYP11A1:15 PROC:24 SLC26A4:47 KCNJ1:77
Left-Sided Heart Failure	0.14104346	113	2.353e-07	7.049e-04	ADRB1:128 ATP2A1:137 GDF15:141 GPX4:185 PTH:193 ALB:217
Coronary Artery Disease	0.05344312	788	5.089e-07	8.317e-04	FGD5:3 PARP12:8 CPE:9 PROM1:13 SLC26A4:47 COL14A1:87
Steatohepatitis	0.06954155	454	4.791e-07	8.317e-04	FFAR1:33 SSGM3:39 SUGP1:58 LATS2:74 ST14:99 CA3:139
Kidney Calculi	0.19339702	53	1.137e-06	1.593e-03	CLDN16:2 AGXT:86 PTH:193 LRRAP1:245 SLC4A1:275 SLC01B1:279
Atherogenesis	0.20830704	43	2.322e-06	2.646e-03	HRH1:226 PTGS2:522 VEGFA:641 ALOX5:707 CNR2:800 HP:809
Metabolic Diseases	0.07382889	350	2.428e-06	2.646e-03	CPE:9 GCDH:12 AGXT:86 LPIN2:111 ARSA:158 GLA:175
Heart Decompensation	0.13970265	93	3.341e-06	2.730e-03	ADRB1:128 ATP2A1:137 GDF15:141 GPX4:185 PTH:193 ALB:217
Hyperammonemia	0.12493661	118	2.898e-06	2.730e-03	GYS2:105 ARSA:158 GLA:175 SMPD1:197 CPT1A:294 NAGS:322
Myocardial Failure	0.13643677	98	3.171e-06	2.730e-03	ADRB1:128 ATP2A1:137 GDF15:141 GPX4:185 PTH:193 ALB:217
Coronary Arteriosclerosis	0.05538642	612	3.774e-06	2.847e-03	CPE:9 PROM1:13 SLC26A4:47 UGT1A6:89 MP2L:103 CPM2:120
Essential Hypertension	0.08811929	225	5.749e-06	4.026e-03	ADRB1:128 PTGFR:140 GDF15:141 GPX4:185 PTH:193 ALB:217
Hypercholesterolemia	0.09122897	208	6.246e-06	4.083e-03	SLC10A2:155 CAPG:156 DAB2:160 CLU:190 ALB:217 NOX1:219
Myocardial Repfusion Injury	0.23146889	31	8.256e-06	4.762e-03	ADRB1:128 ALB:217 CASP12:524 CAT:605 VEGFA:641 BAX:737
Non-alcoholic Fatty Liver Disease	0.07909351	271	8.237e-06	4.762e-03	LATS2:74 ATG7:84 GATA2A:109 MTPP:146 ARSA:158 ABCB11:181
Heart Failure, Right-Sided	0.13081214	95	1.089e-05	5.924e-03	ADRB1:128 ATP2A1:137 GDF15:141 GPX4:185 PTH:193 ALB:217
Acute kidney injury	0.14767745	72	1.508e-05	7.781e-03	CLU:190 ALB:217 AMN:266 HEBX:340 VEGFA:641 B2M:708
Fatty Liver	0.06708644	3	1.698e-05	8.209e-03	FFAR1:33 SSGM3:39 SUGP1:58 LATS2:74 ATG7:84 CA3:139
Hypertriglyceridemia	0.11057626	127	1.758e-05	8.209e-03	ATP6V0A2:62 MTPP:146 SLC01A2:155 SMPD1:197 ALB:217 PHEX:308
Malignant neoplasm of lung	0.03139599	1819	1.936e-05	8.628e-03	BP1FB1:1 IL1R2:41 CPE:9 PROM1:13 VAV1:34 DCAF:38
Drug-Induced Liver Disease	0.08775761	199	2.114e-05	9.012e-03	IL1R2:4 SULT1E1:61 COL14A1:87 CA3:139 ABCB11:181 CLU:190
Hyperlipidemia	0.09175120	180	2.317e-05	9.467e-03	CPE:9 MTPP:146 ARSA:158 GLA:175 PTH:193 ALB:217
Cerebral Infarction	0.08148492	228	2.429e-05	9.528e-03	CALM1:114 ENTDP1:150 NRN1:161 FABP1:204 ISYNA1:263 POU5F1:287
Anemia, Sickle Cell	0.09338964	170	2.816e-05	1.023e-02	UGT1A6:89 CALM1:114 COMT:120 GDF15:141 SPAT1:172 SMPD1:197
Metabolic Syndrome X	0.05799402	453	2.740e-05	1.023e-02	SULT1E1:61 ATG7:84 COMT:120 GDF15:141 ENTDP1:150 ABCB11:181
Hypoferritinemia	0.10833935	125	2.998e-05	1.050e-02	CALM1:114 MTPP:146 ALB:217 SLC01B1:279 CD59:354 HMGCR:357
Obesity	0.0323556	1559	3.669e-05	1.241e-02	CPE:9 SLC23A1:11 PROM1:13 PNLP:26 FABP1:33 SSGM3:39
Anemia, Hemolytic	0.12721179	88	3.862e-05	1.262e-02	SPAT1:172 DMTN:183 KCNN4:209 SLC4A1:275 PRKCD:284 ANK1:335
Kidney Failure, Chronic	0.06482491	344	4.051e-05	1.281e-02	KCNJ1:77 AGXT:86 ENTDP1:150 CAPG:156 GLA:175 CLU:190
Hyperglycemia	0.05881190	413	4.740e-05	1.416e-02	CYP11A1:15 SSGM3:39 ZACN:80 ATG7:84 ADRB1:128 TIPAR:180
Overweight	0.08351301	201	4.767e-05	1.416e-02	FFAR1:33 COMT:120 ALB:217 CPT1A:294 CD59:354 CRY2:393
Acute Kidney Insufficiency	0.14130617	69	5.031e-05	1.451e-02	CLU:190 ALB:217 AMN:266 HEBX:340 VEGFA:641 B2M:708
Dyslipidemia	0.08765138	179	5.537e-05	1.551e-02	MTPP:146 RORA:189 ALB:217 SLC01B1:279 ENPP2:330 SCD:355
Hypercholesterolemia, Familial	0.10403643	125	6.122e-05	1.667e-02	MTPP:146 SLC22A1:184 SLC01B1:279 HMGCR:357 CYP1A7:373 ACAD4:667
Hepatitis, Drug-Induced	0.09053477	164	6.647e-05	1.755e-02	IL1R2:4 SULT1E1:61 COL14A1:87 CA3:139 CLU:190 PKP4:191
Pancreatitis	0.09041095	164	6.802e-05	1.755e-02	PROC:24 PNLP:26 PLAUR:68 PNLPURP2:82 ALB:217 CTRC:237
Chronic Kidney Diseases	0.06750339	296	7.099e-05	1.785e-02	CLDN16:2 IT1G:131 KCNJ1:77 AGXT:86 DAB2:160 SLC22A1:184
Kidney Diseases	0.05174309	508	7.711e-05	1.863e-02	CLDN16:2 SCNN1G:10 IT1G22:31 SULT1E1:61 PLAUR:68 AQP1:131